Patrícia Szabó

Complete List of Publications

Research impact

My research focuses on the development and evaluation of virtual reality (VR)–based serious games and cognitive assessment tools for rehabilitation, assistive technologies, and **digital health applications**. My work combines **human–computer interaction**, immersive technologies, and cognitive psychology to design interactive systems that improve cognitive assessment and **rehabilitation outcomes,** to create accessibility for vulnerable user groups.

One of the main contributions of my research is the adaptation of a neuropsychological assessment methods to immersive environments. In particular, the development of a virtual reality implementation of the Corsi Block-Tapping Test, a widely used spatial memory test in psychology. This work demonstrates how immersive environments can improve engagement and usability while enabling more flexible and automated assessment protocols. [A1, F2, F9, F10]

My research further investigated biofeedback-based VR therapies for rehabilitation and stress reduction. Several VR-based systems were developed including immersive breathing training applications and cognitive training environments designed for patients with post-COVID syndrome [B2], pulmonary disorders, and memory impairments. Results demonstrated that immersive VR environments significantly increase engagement and therapeutic experience. [F4, F6, F7, F8, F12, F13] In comparative experiments with different visualization devices, VR head-mounted displays were rated as the most immersive by 68% of participants, highlighting the importance of immersive visualization technologies in rehabilitation contexts. A comprehensive review of assistive technologies demonstrated the potential of VR, mobile, and wearable technologies as scalable and cost-efficient tools for cognitive support and rehabilitation. [A2]

Through interdisciplinary collaborations with psychologists and healthcare professionals, my work aims to advance evidence-based immersive technologies and other IT solutions that can be integrated into **clinical, educational, and rehabilitation environments**. By combining immersive VR environments, biofeedback, and user-centered design principles, my research contributes to the development of **accessible digital rehabilitation systems that improve both patient engagement and therapeutic outcomes.**

A) Peer-Reviewed: published or accepted for publishing

1. **Szabó, P.**, Pete, K., & Sikné Lányi, C. (2025). A New Path to Recovery: Virtual Reality-Driven Treadmill Exercises. In *Technology for Inclusion and Participation for All: Recent Achievements and Future Directions* (pp. 158–165). Springer. https://doi.org/10.1007/978-3-032-01628-7_20
2. Zsebi, S., **Szabó, P.**, Sikné Lányi, C., & Cserjési, R. (2025). A Full Lifespan Comparison of Virtual Reality Based Cognition Testing on Acceptance, Effectiveness and Satisfaction. In *Technology for Inclusion and Participation for All: Recent Achievements and Future Directions* (pp. 257–264). Springer. https://doi.org/10.1007/978-3-032-01628-7_31
3. **Szabó, P.**, Filotás, P., Sikné Lányi, C., Zsebi, S., & Cserjési, R. (2024). Virtual reality implementation of the Corsi test and pilot study on acceptance. *Software Impacts*, 21, 100693. https://doi.org/10.1016/j.simpa.2024.100693
4. Losonci, L., Sikné Lányi, C., & **Szabó, P.** (2024). Modelling realistic avatars for the "P-game". In *Proceedings of the 15th International Conference on Disability, Virtual Reality and Associated Technologies (ICDVRAT 2024)*(pp. 126–128).
5. **Szabó, P.**, & Sikné Lányi, C. (2024). Design virtual reality games that instruct proper breathing techniques with dynamically changing virtual environment. In *Proceedings of the 15th International Conference on Disability, Virtual Reality and Associated Technologies (ICDVRAT 2024)* (pp. 166–167).
6. **Szabó, P.**, & Sikné Lányi, C. (2024). User-friendly serious game design for diabetic preschool children. In *Future Perspective on Accessibility, Assistive Technology and (e)Inclusion – ICCHP 2024* (pp. 65–69).
7. **Szabó, P.**, & Sikné Lányi, C. (2024). Mesterséges intelligencia szerepe a 3D memóriajáték eredményességében [The role of artificial intelligence in the effectiveness of a 3D memory game]. In *A digitalizáció és a mesterséges intelligencia hatása az egészségügy jövőjére*.
8. **Szabó, P.**, Ara, J., Halmosi, B., Sikné Lányi, C., & Guzsvinecz, T. (2023). Technologies designed to assist individuals with cognitive impairments. *Sustainability*, 15, 13490. https://doi.org/10.3390/su151813490
9. **Szabó, P.**, Sikné Lányi, C., Filotás, P., & Cserjési, R. (2023). Virtual reality-based application for rehabilitation: Corsi-test. In *2nd IEEE International Conference on Cognitive Aspects of Virtual Reality – cVR 2023* (pp. 29–32). https://doi.org/10.1109/CVR58941.2023.10395839
10. Zsebi, S., **Szabó, P.**, Filotás, D., Sikné Lányi, C., & Cserjési, R. (2023). Enhancing neuropsychological assessment through virtual reality: A pilot study of the Corsi-Block tapping task. In *2nd IEEE International Conference on Cognitive Aspects of Virtual Reality – cVR 2023* (pp. 115–118).

11. **Szabó, P.**, Baranyi, P. Z., & Sikné Lányi, C. (2023). DOS-Windows-Virtual desktop. In *2nd IEEE International Conference on Cognitive Aspects of Virtual Reality – cVR 2023* (pp. 33–36).
12. **Szabó, P.**, & Sikné Lányi, C. (2023). Virtual reality based serious games for memory skill improvement. In *XXXVI. Neumann Kollokvium*.
13. Sikné Lányi, C., Guzsvinecz, T., Tálas, M., Halmosi, B., **Szabó, P.**, & Haneklaus, N. (2023). Játékra fel! – Virtuális valóság alapú „Negotiation Game". [Let's play! – A virtual reality-based "Negotiation Game".] In *29th Multimedia in Education Conference*.
14. **Szabó, P.**, & Sikné Lányi, C. (2023). A digitális és hagyományos oktatás, generációs különbségek. [Digital and traditional education, generational differences.] In *29th Multimedia in Education Conference*.
15. Sikné Lányi, C., & **Szabó, P.** (2023). Developing an Android-based game for children with blindness or low vision. In *AAATE 2023 Book of Abstracts*.
16. **Szabó, P.**, & Sikné Lányi, C. (2022). Android alapú alkalmazás tervezése 1-es típusú cukorbeteg gyermekek tanítására [Designing an Android-based application to teach children with type 1 diabetes]. In *XXXV. Neumann Kollokvium*.
17. **Szabó, P.**, Sikné Lányi, C., Schalbert, J., Kretz, Z., & Cserjési, R. (2022). „Légzés félelem nélkül" – Virtuális valóság alapú légzés javító játék tervezése post és long-Covid szindrómában szenvedő betegek rehabilitációjára. ["Breathing Without Fear" – Designing a virtual reality-based breathing improvement game for the rehabilitation of patients suffering from post-COVID and long-COVID syndrome.] In *XXXV. Neumann Kollokvium*.
18. Bodor, B., **Szabó, P.**, Mogánné Tölgyesy, S., & Sikné Lányi, C. (2019). Rehabilitációs játék készítése tabletre. In *XXXII. Neumann Kollokvium*.

B) Non Peer-Reviewed: published

1. Nils, H., László, H., Hendrik, B., Tzong-Ru, L., Matúš, M., Mijalche, S., **Szabó, P.**, Guzsvinecz, T., & Sikné Lányi, C. (2025). Negotiating virtually and face-to-face: First experience from a serious game conducted in person and via smartphone application. *Research Square (Preprint)*. https://doi.org/10.21203/rs.3.rs-6110762/v1
2. **Szabó, P.** (2023). *How can we use VR to fight post-COVID syndrome?* YouTube. https://www.youtube.com/watch?v=_j427Z0SIMc

F) Further published contributions to the Scientific Community

1. **Szabó, P.**, & Sikné Lányi, C. (2024). Immersive VR games for cognitive enhancement. In *womENcourage™ 2024: Responsible Computing for Gender Equality*. ACM Celebration of Women in Computing.
2. **Szabó, P.** (2024). Immersive VR games for cognitive enhancement. In *XXVII. Tavaszi Szél Konferencia 2024 – Abstract volume*.
3. **Szabó, P.** (2023). Revolutionizing rehabilitation through VR-based serious games. In *Technicity Doctoral Workshop Conference – MOME*.
4. **Szabó, P.**, Bodor, B., & Sikné Lányi, C. (2019). Tabletre játék tervezése. [Designing games for tablets.] In *25th Multimedia in Education Conference*.
5. Bodor, B., **Szabó, P.**, & Sikné Lányi, C. (2019). Re-Creation, an Android game. In *Pannonian Conference on Advances in Information Technology (PCIT 2019)*.